

Population-Scale Decomposition of TLE B* Variability Across 19,671 LEO Objects

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Abstract

Two-line element (TLE) sets encode atmospheric drag via the B* parameter, yet the temporal variability of B* conflates atmospheric density model errors, spacecraft maneuvers, solar radiation pressure, and physical drag coefficient (Cd) changes. Using 43.9 million TLEs spanning 19,671 low Earth orbit objects — the largest such analysis to date — we decompose B* variability into three components through population self-calibration. A shared atmospheric signal, removed via common-mode subtraction across altitude-month bins computed exclusively from non-constellation objects, explains 42% of the total variance. A mega-constellation maneuver signature, identified by orbital parameter fingerprinting of 11,689 Starlink, Planet, and related satellites, produces median corrected CV = 1.13 — more than double the non-constellation value of 0.48. After filtering to 2,239 non-constellation, non-maneuvering objects and applying outlier removal and linear detrending, the residual coefficient of variation at 100–200 km altitude is 0.255 (N = 109), providing an upper bound on combined Cd variability and SGP4 fitting noise that is consistent with physical Cd variability reported in the literature (CV = 0.10–0.30). Above 300 km the residual CV plateaus at 0.38–0.42, consistent with the growing contribution of solar radiation pressure to B*. These results quantify the precision limits of TLE-derived drag analysis and provide empirical variance budgets for operational collision avoidance and reentry prediction.

1. Introduction

The two-line element (TLE) format has served as the primary public interface to the US Space Surveillance Network catalog for over five decades. Each TLE encodes a fitted B* drag parameter that absorbs atmospheric drag effects into the SGP4/SDP4 analytical propagator framework (Hoots & Roehrich, 1980; Vallado et al., 2006). B* is formally defined as

$$B^* = C_d A \rho_0 / (2m)$$

where C_d is the drag coefficient, A is the cross-sectional area, m is the mass, and ρ_0 is a reference atmospheric density. In practice, B* is a fitted parameter from the orbit determination process, absorbing not only drag but also solar radiation pressure (SRP), unmodeled perturbations, and fitting residuals. The ratio B^*/ρ is therefore not a clean measure of $C_d \cdot A / (2m)$ but rather an effective ballistic parameter contaminated by cross-model density errors between the fitting

model used by the 18th Space Defense Squadron and any external density model applied by the analyst.

Understanding B^* variability matters for three operational reasons. First, collision avoidance systems rely on propagated orbits whose accuracy depends on drag modeling, and B^* scatter directly translates to propagation error budgets. Second, atmospheric density estimation from satellite drag requires separating true density variations from C_d and A/m changes — a degeneracy that has limited TLE-based density retrieval (Emmert, 2015). Third, the explosive growth of mega-constellations has introduced thousands of actively maneuvering objects whose B^* behavior differs fundamentally from legacy debris populations.

Previous work on satellite drag coefficients has focused on individual objects with known physical properties. Pilinski et al. (2013) computed C_d variations of 10–30% for simple geometries across different atmospheric compositions using DSMC methods. Mehta et al. (2014) demonstrated altitude- and solar-activity-dependent C_d variations for spherical and flat-plate geometries. Doornbos (2012) provided detailed C_d variability measurements from accelerometer data on dedicated missions, establishing ground truth of $C_d \approx 2.0$ – 2.3 with approximately 10–15% variation for known geometries. On the density retrieval side, Emmert (2015) used population-averaged TLE-derived densities to study long-term thermospheric trends, explicitly noting B^* fitting noise as a limitation requiring population averaging. Licata et al. (2021) developed machine-learned thermospheric density models using TLE-derived data with explicit uncertainty quantification.

No prior study has attempted a population-scale decomposition of B^* variability across the full LEO catalog. This work fills that gap using 43.9 million TLEs from 19,671 objects, applying a self-calibration method that requires no external reference satellites.

2. Data

2.1 TLE Catalog

The input dataset consists of 43.9 million two-line element sets parsed from a 6.3 GB archive containing the full publicly available TLE catalog as of May 2026. The file contains TLEs in two-line format with interleaved blank lines and two NORAD ID formatting variants (space-padded and zero-padded). A streaming parser identified TLE line pairs by checking the line-type digit at position 0, extracting NORAD catalog numbers from positions 2–7 with whitespace tolerance.

After parsing, 36.9 million TLEs passed the LEO filter (apogee < 1000 km), spanning 22,327 unique objects. Epoch-level deduplication at minute resolution removed 5.9 million duplicate entries arising from the two NORAD ID format variants, yielding 31.0 million unique LEO TLE records.

2.2 Object Selection

Objects with fewer than 20 unique epochs were excluded, leaving 19,671 objects for analysis. The

median object has 705 epochs spanning 539 days. The maximum epoch count is 13,508. The altitude distribution peaks at 502 km (median perigee), with an interquartile range of 456–553 km, reflecting the dominance of mega-constellations in the contemporary LEO population.

2.3 Mega-Constellation Identification

Mega-constellation satellites were identified using a combination of NORAD catalog number ranges and orbital parameter fingerprints rather than name-based matching. Starlink satellites were identified by NORAD ID $> 44,000$ combined with inclination within 2° of their operational shells (53° , 43° , 70° , and 97.6°) at perigee altitudes between 300 and 600 km. Planet Dove satellites were identified at inclination $97.4^\circ \pm 1.5^\circ$ in the 400–550 km altitude range with NORAD ID $> 40,000$. This classification identified 11,689 constellation satellites in the final dataset: 5,327 Starlink at 53° inclination, 2,150 Starlink at 97.6° , 2,052 Planet, 1,365 Starlink at 43° , 506 Starlink at 70° , and 299 probable Starlink. The remaining 7,982 objects constitute the non-constellation population.

3. Method

3.1 Ballistic Parameter Time Series

For each object at each epoch, the log-ratio $\log(B^*/\rho_{\text{model}})$ was computed, where ρ_{model} is an atmospheric density estimate at the perigee altitude. The density model uses a piecewise exponential profile with base densities and scale heights calibrated to NRLMSISE-00 at moderate solar activity (Picone et al., 2002), with a solar-cycle correction applied via a sinusoidal F10.7 proxy. The solar correction uses an altitude-dependent power law, $\rho \propto (F10.7/150)^\alpha$, with α ranging from 0.5 at 150 km to 2.8 above 800 km.

We note that this density model is an analytical approximation and differs from the internal model used by the 18th Space Defense Squadron in the TLE fitting process. The ratio B^*/ρ_{model} therefore contains a cross-model density error term that varies with altitude and solar conditions. This systematic error is partially but not fully removed by the common-mode subtraction described in Section 3.3.

3.2 Outlier Removal

Each object's $\log(B^*/\rho)$ time series was filtered using a Median Absolute Deviation (MAD) criterion applied globally to the object's full time series. Points deviating by more than $3 \times \text{MAD}$ from the object's median were removed, with a minimum threshold of 0.3 in log-space to avoid over-filtering stable objects. This step removed 1,966,004 data points (approximately 7% of the clean dataset), targeting maneuver artifacts, SGP4 fitting glitches, and attitude-change transients.

3.3 Population Self-Calibration

The self-calibration method exploits the fact that objects at the same altitude and observed

during the same month share the same atmospheric density model error. Critically, only non-constellation objects contribute to the common-mode computation. Including mega-constellation satellites would contaminate the atmospheric correction with maneuver-induced B^* variability, as constellation satellites undergo station-keeping, orbit-raising, and deorbit maneuvers that produce B^* excursions unrelated to atmospheric density.

Each surviving data point from the 7,984 non-constellation objects was assigned to an (altitude bin, month) cell, using 25-km altitude bins and calendar-month time bins. For each cell, the population median of $\log(B^*/\rho)$ was computed as the common-mode signal. This yielded 2,032 common-mode bins with a median population of 1,760 non-constellation objects per bin. Bins with fewer than 5 contributing objects were excluded from the correction.

For each object (both constellation and non-constellation), the common-mode was subtracted from its $\log(B^*/\rho)$ time series at each epoch using the corresponding (altitude, month) bin. This removes the component of B^* variability shared across co-altitude objects in the same time period — primarily atmospheric density model error.

3.4 Linear Detrending

After common-mode subtraction, each object's residual time series was linearly detrended against time. This removes secular changes from orbit decay (changing altitude and thus density model bias), gradual solar panel degradation (changing A/m), and slow attitude evolution. Only the scatter around the linear trend contributes to the reported variability. Objects with fewer than 10 valid residuals or time spans shorter than 30 days were not detrended.

3.5 Variability Metric and Classification

The corrected coefficient of variation (CV) for each object is defined as the interquartile range of the detrended, common-mode-subtracted $\log(B^*/\rho)$ residuals. In log-space, IQR is equivalent to a robust fractional CV for moderate variability.

Objects were classified as "stable" (pre-correction $\log\text{-IQR} < 0.5$) or "erratic" based on their raw variability before common-mode subtraction. The "clean subset" — 2,239 objects that are both non-constellation and stable — represents the population most likely to exhibit physical Cd variability rather than maneuver- or attitude-induced B^* changes. We note that this selection on raw variability introduces a threshold dependence: the absolute CV values of the clean subset depend on the 0.5 threshold, though the altitude trend within the clean subset is robust to threshold choice.

4. Results

4.1 Variance Budget

The common-mode subtraction plus linear detrending together explain 42% of the total B^* variance across all 19,671 objects. The atmospheric common-mode alone (before detrending)

accounts for approximately 24% — substantially more than the 9% obtained when constellation objects were included in the common-mode computation (see Section 5.2). The remaining 58% of variance is object-specific: a combination of maneuvers, attitude changes, SRP variations, SGP4 fitting noise, and physical Cd variability.

4.2 Constellation vs. Non-Constellation

The 11,689 constellation satellites have a corrected median CV of 1.13, more than double the non-constellation value of 0.48. This separation is expected: constellation satellites undergo routine station-keeping maneuvers, orbit-raising after deployment, and end-of-life deorbit burns, all of which produce large B* excursions that persist even after outlier removal. The constellation population constitutes 59% of the LEO catalog by object count, meaning that any bulk analysis of TLE B* variability that does not separate constellation from non-constellation objects will be dominated by maneuver-induced variability unrelated to atmospheric drag.

4.3 Altitude Dependence of Residual Variability

The altitude profile of corrected CV for the 2,239 clean-subset objects (non-constellation, stable) is as follows:

Altitude Band (km)	Median CV	N
100-200	0.255	109
200-300	0.344	238
300-400	0.410	600
400-500	0.419	857
500-600	0.380	373
600-800	0.357	62

The lowest-altitude band (100–200 km, N = 109) shows the smallest residual variability at CV = 0.255. This value represents an upper bound on the combined contribution of physical Cd variability and SGP4 fitting noise. It is consistent with the Cd variability range of CV = 0.10–0.30 reported by Pilinski et al. (2013) for objects in the free molecular flow regime, though the Cd contribution cannot be isolated from the SGP4 noise floor without independent validation.

The increase in CV from 0.255 at 100–200 km to a plateau of 0.38–0.42 at 300–500 km is consistent with the growing contribution of solar radiation pressure to B*. At altitudes above 400 km, atmospheric drag weakens while SRP — which varies with sun angle, spacecraft attitude, and Earth shadow geometry — becomes a comparable or dominant non-gravitational force. The

B^* fitting process absorbs SRP into the drag parameter, introducing variability unrelated to either atmospheric density or C_d .

The decrease at 500–600 km and 600–800 km likely reflects a selection effect: objects maintaining altitude above 500 km without active propulsion tend to have low area-to-mass ratios (compact, massive bodies), reducing both drag and SRP variability. Sample sizes at these altitudes ($N = 373$ and $N = 62$) limit the statistical power of this observation.

4.4 Stable vs. Erratic Objects

The 3,567 stable objects (raw log-IQR < 0.5) have a corrected median CV of 0.40 with an IQR of 0.32–0.47. The 16,104 erratic objects have a corrected median CV of 0.74. This $1.9\times$ ratio persists after common-mode subtraction and detrending, confirming that the stable/erratic distinction reflects genuine per-object behavioral differences (passive debris vs. active spacecraft) rather than atmospheric conditions.

5. Discussion

5.1 What B^* Actually Measures

The central finding of this work is that TLE B^* variability is predominantly object-specific. Even with an improved common-mode computed exclusively from non-constellation objects, shared atmospheric error accounts for only 42% of the variance. The remaining 58% is driven by maneuvers (for constellation and active satellites), SRP (increasing with altitude), SGP4 fitting noise, and physical C_d variability.

This has direct implications for studies that treat B^* as a density probe: the signal-to-noise ratio for atmospheric density retrieval from individual TLEs is poor. Population-level averaging over many co-altitude, non-maneuvering objects — as performed by Emmert (2015) — is necessary to suppress the per-object noise. Our results quantify this noise floor for the first time at population scale.

5.2 The Constellation-Contaminated Common-Mode

An important methodological finding is the sensitivity of the common-mode to population composition. When constellation satellites were included in the common-mode computation (not shown in final results), the variance explained dropped from 42% to 26%, and the clean-subset correction worsened (negative reduction of -10%), meaning the common-mode subtraction was adding noise to clean objects. This occurred because Starlink satellites, which constitute over 5,000 objects at 53° inclination and 530–560 km altitude, dominated the common-mode in those altitude-month bins. Their maneuver-induced B^* variations were treated as "atmospheric signal" and subtracted from non-constellation objects, corrupting the correction.

This finding has implications for any future TLE-based density study: the common-mode or population-average approach to density retrieval will be increasingly degraded as mega-constellations grow to dominate the LEO catalog, unless constellation objects are explicitly excluded from the averaging.

5.3 Interpreting the Low-Altitude Residual

The $CV = 0.255$ at 100–200 km altitude requires careful interpretation. This value is an upper bound on physical C_d variability because it includes contributions from: SGP4 orbit determination fitting noise, unmodeled perturbations not captured by B^* , cross-model density errors between the 18th SDS fitting model and our analytical density approximation, and actual C_d variation from changes in atmospheric composition, temperature, and object orientation relative to the flow.

Without independent calibration — ideally from precision orbit determination of objects with known geometry, or from accelerometer-equipped satellites — we cannot decompose this residual further. The consistency with Pilinski et al. (2013) and Doornbos (2012) is suggestive but not conclusive.

The $N = 109$ sample at 100–200 km is composed predominantly of decaying objects with short remaining lifetimes and rapidly evolving perigees. Their B^* values benefit from high atmospheric density (10^2 – $10^3\times$ higher than at 500 km), where drag strongly dominates SRP and the exponential density model is most accurate. The low CV at these altitudes therefore partially reflects better density modeling rather than lower C_d variability.

5.4 Limitations

The density model is a smooth analytical approximation with a sinusoidal F10.7 proxy that does not capture short-term solar variability (27-day rotation, geomagnetic storms) or cycle-to-cycle amplitude differences. Using actual NRLMSISE-00 or JB2008 outputs (Bowman et al., 2008) with measured solar and geomagnetic indices would reduce the common-mode residual but is unlikely to change the per-object component substantially.

The constellation identification is heuristic and based on orbital parameter fingerprinting. While robust for the dominant Starlink shells, it may misclassify individual objects. The "stable" threshold of raw $\log\text{-IQR} < 0.5$ is arbitrary; the absolute CV values of the clean subset are threshold-dependent, though the altitude trend is robust.

The MAD-based outlier filter is applied globally to each object's time series rather than in a rolling window, which may inadequately handle objects with genuine long-term B^* shifts (e.g., attitude changes at mission midpoint).

The absence of known- C_d calibration satellites (STELLA, Starlette, AJISAI, LAGEOS) from the TLE archive prevented direct calibration of the B^*/ρ ratio against objects with established $C_d \approx 2.2$. Future work incorporating these objects from Space Track historical archives would enable absolute C_d estimation.

5.5 Implications for Operations

For collision avoidance, the variance budget quantifies expected B^* scatter as a function of object type and altitude, enabling orbit propagation uncertainty models to be tuned empirically. The finding that 59% of the LEO population consists of constellation satellites with $CV > 1.0$ highlights the need for maneuver-aware propagation for the majority of conjunction screening events.

For reentry prediction, the low-altitude $CV \approx 0.255$ provides an empirically grounded upper bound on C_d uncertainty that can inform Monte Carlo uncertainty propagation for uncontrolled reentries.

6. Conclusions

We have performed the first population-scale decomposition of TLE B^* variability using 43.9 million TLEs from 19,671 LEO objects. The key findings are:

1. The shared atmospheric signal accounts for 42% of B^* variance when the common-mode is computed exclusively from non-constellation objects, dropping to 26% when constellation satellites contaminate the computation. This establishes that population-based density retrieval from TLEs requires explicit exclusion of mega-constellation objects.
2. Mega-constellation maneuvers dominate B^* variability for 59% of the current LEO population, with corrected $CV = 1.13$ compared to 0.48 for non-constellation objects. This quantifies the "constellation contamination" of the TLE catalog for drag analysis.
3. The residual per-object variability for non-maneuvering objects at 100–200 km altitude is $CV = 0.255$ ($N = 109$), providing a population-level upper bound on combined C_d variability and SGP4 fitting noise. This is consistent with but cannot be distinguished from the physical C_d variability range of 0.10–0.30 reported in the literature.
4. The altitude dependence of residual variability — increasing from $CV = 0.255$ at 100–200 km to a plateau of 0.38–0.42 at 300–500 km — is consistent with the growing contribution of solar radiation pressure at altitudes where atmospheric drag weakens.

The self-calibration method introduced here — common-mode subtraction across altitude–time bins using only non-constellation objects — is applicable to any TLE archive and provides a framework for ongoing monitoring of B^* variability as the LEO population continues to evolve.

Data Availability

The analysis code and derived data products (per-object CSV with corrected variability metrics) are available from the author upon request. The input TLE data were obtained from public Space Surveillance Network archives.

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